

Shree Pandit

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EDUCATION

Maulana Azad National Institute of Technology, Bhopal	Expected May 2028
B.Tech - Computer Science and Engineering	CGPA: 7.40
The Himalaya Academy, Dewas	2023
Madhya Pradesh Board of Secondary Education – Class XII	Percentage: 85.2%
Carmel High School, Gadchiroli	2021
Central Board of Secondary Education – Class X	Percentage: 93.8%

WORK EXPERIENCE

Data Engineering Intern	May 2026 – July 2026
Transform Rural India Foundation (TRIF)	
- Centralized 186,000+ unstructured survey responses into 7 structured fact tables, resolving all schema inconsistencies and encoding errors, by architecting a grain-aware star-schema ETL pipeline in Python (Pandas, NumPy). - Accelerated KPI reporting from weekly manual exports to near-real-time across 2,000+ Gram Panchayats in 5 states, by designing a multi-page Power BI dashboard tracking 150+ metrics. - Cut manual report generation time by 80% for BLCC Fact Sheets, by automating end-to-end web scraping workflows via Python (Selenium).	

PROJECTS

A.R.G.U.S C++, Python, FastAPI, React, Tailwind, ZeroMQ, TimescaleDB, Redis	GitHub
AI-driven Recognition Gateway for Unidirectional IP Security in Critical Infrastructure Networks.	
- Enabled passive threat detection on air-gapped, unidirectional networks with zero return-path capability, by engineering a C++ DPI engine to extract flow-level security features (JA3 fingerprints, DNS entropy, host fan-out profiling). - Bridged real-time ML inference into a live SOC dashboard with zero data loss between detection and visualization layers, by architecting a ZeroMQ PUB/SUB pipeline and a schema-translating consumer service. - Delivered alert broadcasting with automatic Redis-failover recovery, by building a FastAPI backend with Redis Pub/Sub and WebSocket fan-out, backed by a fail-closed PostgreSQL/TimescaleDB persistence layer.	
Auralis C++, Python, FastAPI, React.js, Npcap, WebSockets	GitHub
Real-time network traffic diagnostic instrument featuring cross-platform C++/Python IPC.	
- Achieved sub-10ms parsing of raw L2/L4 network frames, by engineering a C++ capture engine (libpcap/Npcap) streaming 5-tuple flow data via a custom TCP IPC bridge. - Delivered zero-latency metrics with no database persistence, by architecting an async Python (FastAPI) aggregation layer using a 300-second in-memory ring buffer for flow tracking. - Rendered live bandwidth and connection states at 1-second intervals, by developing a React/Tailwind dashboard with an SVG oscilloscope trace and a WebSocket-driven patch-panel visualization.	
NITBFreshers Study Portal JavaScript, Tailwind CSS, PHP, MySQL	GitHub
Full-stack academic resource management system for MANIT students.	
- Sustained 1,000+ concurrent users (95% batch adoption) with zero downtime, by deploying a multi-instance PHP/MySQL backend and optimizing database queries for rapid resource filtering. - Cut document load times by 40%, by replacing synchronous page reloads with asynchronous Fetch API calls and integrating PDF.js for seamless in-browser rendering. - Secured 100% of academic assets against unauthorized access, by enforcing robust session management using 64-character HttpOnly tokens and parameterized database operations.	

RESEARCH EXPERIENCE

End-to-End Geotechnical Data Extraction Framework	August 2026 – Present
Guided by Dr. Kishan Dharavat, Dept. of Civil Engineering, MANIT Bhopal	
- Achieved a 0.98 F1-score in classifying unstructured geotechnical PDFs, by engineering a hybrid pipeline combining an EfficientNet-b0 CNN with a fallback text-searching algorithm. - Automated complex layout parsing with a 0.95 mAP (mean Average Precision) on the evaluation set, by implementing a Faster R-CNN for spatial bounding-box detection feeding a custom rule-based engine to generate structured header-value pairs.	

OPEN SOURCE CONTRIBUTIONS

EDMS – Endpoint Data Management System Rust, Axum, React.js	GitHub
Open-source project by HashedTokens – Core Contributor.	
- Synchronized the disconnected React frontend and Rust backend, eliminating 100% of fabricated client-side state, by wiring the UI directly to live REST APIs and WebSocket data streams. - Restored full end-to-end endpoint testing functionality, resolving critical IPC fallback failure by patching Axum dispatcher bugs and migrating HTTP client logic to the active child process.	

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, PHP
Web & Frameworks: React.js, Next.js, FastAPI, Node.js, Express.js, Tailwind CSS
Data Libraries and Machine Learning: CNN (EfficientNet), Faster R-CNN, NumPy, Pandas, Scikit-learn
Databases: PostgreSQL, MySQL, MongoDB, TimescaleDB, Redis
Tools & Platforms: Git, Docker, ZeroMQ, Microsoft Power BI
CS Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Artificial Intelligence, Data Science

ACHIEVEMENTS

2nd Runner-Up at Kinetic Coding Challenge by IIPE Visakhapatnam. - National Finalist at Shaastra Programming Contest (IIT Madras) and Code Wars, Prometeo' 26 (IIT Jodhpur). - Solved 1,500+ problems in 150+ contests, earning Guardian (LeetCode), Expert (Codeforces), and 4-Star (CodeChef) titles.	
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POSITION OF RESPONSIBILITY

Corporal, 12 MP BN NCC, MANIT Bhopal	August 2024 – Present
Team Management Head, ISTE SC MANIT	February 2025 – Present